

PROCUREMENT INTELLIGENCE MASTERSHEET

OPERATIONS, RISK & FUTURE AI

Part 2 of 2

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KPI Pillars

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Sourcing Steps

6

Risk Flags

6

AI Tools

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Frameworks

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Maturity Checks

SECTION 6 · THE COMPLETE PROCUREMENT KPI DASHBOARD (6 PILLARS · MODERN CPO STANDARD)

1 · Cost & Value KPIs

Total Cost of Ownership (TCO)

Purchase price + quality + delivery + inventory + end-of-life cost

Spend Under Management (%)

Target: >85%. Below 70% = maverick spend crisis

Cost Savings vs Cost Avoidance

Savings = actual reduction · Avoidance = price increase prevented

Price Variance by Category

Budget vs actual · Flag >5% variance for review

2 · Supplier Performance KPIs

On-Time Delivery Rate (OTDR)

Target: >95%. Below 90% = SLA breach trigger

Supplier Quality / Defect Rate

PPM defects · Target: <500 PPM

SLA Compliance Rate

% of agreed terms met across lead time, quality, pricing

Supplier Scorecard (Quarterly)

Quality 40% + Delivery 30% + Cost 20% + ESG 10%

3 · Procurement Efficiency KPIs

PO Cycle Time

Requisition to PO · Best-in-class: <24 hours automated

Cost per Purchase Order

Manual: █800–2,500 · Automated: █80–250 · AI: <█50

Touchless Procurement Rate

Target: >60% of POs processed without human touch

Digital Adoption Rate

% via digital/e-catalog vs manual requisition

4 · Risk & Resilience KPIs

Single-Source Dependency Ratio

Target: <20% of critical spend on single supplier

Supplier Risk Exposure Index

Financial health + geopolitical + ESG + operational composite

Supply Disruption Rate

Disruptions per quarter · Track recovery time (MTTR)

Business Continuity Score

% of critical categories with pre-qualified alternate vendor

5 · ESG & Sustainability KPIs

% Spend with Sustainable Suppliers

Target 2025: >30%. Target 2030: >60%

Carbon Footprint per Supplier

Scope 3 emissions tracking · EU CSRD regulatory requirement

Diversity Supplier Spend (%)

MSMEs, women-owned, minority-owned · ESG investor scrutiny

Waste Reduction Metrics

Packaging waste, returns rate, end-of-life recovery % per category

6 · Digital Procurement KPIs

AI-Driven Spend Coverage

% of spend categories with AI should-cost models active

Forecast Accuracy (%)

Target >90% · AI models beat human forecasting by >15%

Data Quality Score

Clean, standardised data across ERP + POs + invoices + suppliers

Real-time Dashboard Usage

% of decisions from live data vs monthly reports

SECTION 7 · STRATEGIC SOURCING PROCESS (A.T. KEARNEY 7-STEP · GLOBAL STANDARD)

1. Spend Analysis Analyse spend data, classify categories, identify savings opportunities	2. Category Profiling Assess requirements, map supply market, identify Kraljic position	3. Market Research Study supply market, benchmark prices, map manufacturer landscape	4. Strategy Dev. Develop sourcing strategy, target pricing, negotiation approach	5. Supplier Engage. Conduct RFx process, evaluate bids on TCO basis	6. Negotiation Negotiate & finalise contracts with BATNA-backed strategy	7. Implement & Manage Optimise, monitor KPIs, continuous improvement loop
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■ SECTION 8 · SUPPLY RISK MATRIX SECTION 9 · AI TOOLS ECOSYSTEM 2025

HIGH — Single-Source Dependency

Pre-qualify second vendor always, before need arises. KPI: single-source ratio <20% of critical spend. Toyota Keiretsu proof: survived 2011 earthquake faster than all competitors.

HIGH — Price Spike / Commodity Surge

Forward buy or activate index-linked contracts when commodity index rises >8%. Track LME/MCX/NYMEX. IKEA locked timber costs for decades via forest ownership.

MEDIUM — Vendor Financial Distress

Credit check + payment term buffer at onboarding. Monitor MCA filings quarterly. Flag >30-day payment delays = early distress indicator.

MEDIUM — Geopolitical / Trade Disruption

Diversify sourcing across multiple regions proactively. China+1 strategy for critical categories. Near-shore beats far-shore on total disruption cost.

LOW — Quality / Compliance Failure

On-site audit before volume scale-up. ISO 9001:2015 / BSCI / CTPAT certification mandate. Toyota Jidoka: stop-at-defect costs 1/100th of end-of-line detection.

LOW — Lead-Time Inflation

Safety stock trigger fires automatic alert. Safety Stock = $Z\text{-score} \times \sigma(\text{lead time demand})$. Kanban pull prevents both stockout and overstock simultaneously.

Claude AI (Anthropic)

Should-cost modelling · Market mapping · RFQ drafting · Bid analysis · Negotiation scripts · Risk monitoring · 7-step Balamurugan Framework

Odoo / SAP / Tally ERP

PO tracking · Invoice matching (3-way) · Vendor SLA monitoring · Escalation alerts · Digital PO dashboard (Odoo customised — 4 years active)

AI Demand Forecasting

Walmart model: 500M+ product-store combos weekly. Weather + events + social trends. Accuracy >90% AI vs 75% human forecasting.

Dynamic Pricing Monitor

Amazon: 2.5M price changes/day. Buyer tools: CamelCamelCamel · Keepa · price history. Buy on downward trend — overstock = price floor.

RFID / IoT Inventory

Walmart/Zara model: real-time stock tracking. Every item tagged. Auto-replenishment. Centralised hub = 48hr global restock. 98% OTIF target.

Route Optimization AI

Google Maps API + ML reduces fleet fuel 15–25%. Cross-docking: 65% less handling. Milk-run routing for multi-stop pickups. Load optimization.

■ SECTION 10 · OPTIMUM INVENTORY MODEL (SCIENTIFIC FORMULAS)

EOQ = $\text{SQRT}(2 \times \text{Demand} \times \text{Ordering Cost} / \text{Holding Cost per Unit})$ **Safety Stock** = $Z\text{-score} \times \sigma(\text{lead time demand})$ | $Z = 1.65$ for 95% service level **Reorder Point** = $(\text{Avg Daily Demand} \times \text{Lead Time}) + \text{Safety Stock}$ **Inventory Turnover** = $\text{COGS} / \text{Avg Inventory}$ | Target: >8x annually **Days Inventory Outstanding (DIO)** = $365 / \text{Inventory Turnover}$ | Target: <45 days **ABC-XYZ Matrix**: A-items (80% value, 20% SKUs) = daily monitoring · B = weekly · C = monthly/min-max auto. X = predictable demand, Y = variable, Z = erratic. AX = most critical. CZ = consider elimination.

■ SECTION 11 · 7 ESSENTIAL PROCUREMENT FRAMEWORKS (FIELD-VERIFIED)

1. Kraljic Matrix	4 quadrants: Routine (automate), Leverage (competitive bid), Bottleneck (secure supply), Strategic (partner). Maps spend vs risk.	
2. TCO Model	Purchase price = 30–60% of true cost. Add: quality, logistics, inventory carrying (25–30% of purchase price), disposal. Toyota's 5S reduces waste.	
3. Make vs Buy	Make if: core competency, IP sensitive, volume justifies fixed cost. Buy if: market competition, supplier has scale advantage, capital intensive.	
4. Porter's 5 Forces	Map buyer power vs supplier power, threat of substitutes, new entrants, rivalry. High supplier concentration = risk.	

5. Carter's 10 Cs	Competency · Capacity · Commitment · Control · Cash · Cost · Close supplier · Collaborate · Clean (ESG) · Communication.	On-site supplier · On-board · Clean (ESG) · Communication.
6. SWOT + PESTEL	SWOT for supplier health assessment. PESTEL for supply market analysis.	Annual · Political, Economic, Social, Technology, Environmental · Technology, Environmental
7. BATNA Negotiation	Best Alternative to Negotiated Agreement. Always define your BATNA before entering negotiation room. The party	Every commercial negotiation

■ SECTION 12 · PROCUREMENT EVOLUTION ROADMAP — PAST TO FUTURE

Manual / Relationship	Pre-2000	Paper POs, telephone RFQs. Information asymmetry maximum. Price governed by relationship,
ERP / eSourcing	2000–2015	SAP/Oracle adoption. E-auctions. Spend analytics. Category management formalised. RFID beg
Analytics / Cloud	2015–2023	Real-time dashboards. Supplier portals (Retail Link model). AI demand forecasting. Dynamic pric
AI-Powered (Current)	2024–2025	Claude AI for should-cost + negotiation scripts — Balamurugan Framework active. Autonomous
Autonomous/Agentic	2026–2030	Fully autonomous agents place orders, monitor quality, renegotiate on price triggers. Blockchain

■ SECTION 13 · 12-POINT PROCUREMENT MATURITY CHECKLIST

	✓ Single-source dependency <20% for all critical categories
✓ Strategy linked to business goals — procurement at Board level	✓ RFID/IoT real-time inventory tracking — no manual count
✓ KPIs track value beyond savings — impact, not just cost	✓ AI-driven demand forecasting accuracy >90% per category
✓ Spend under management >85% — zero maverick spend	✓ ESG supplier scorecard active — carbon footprint tracked
✓ Suppliers segmented strategically using Kraljic Matrix	✓ Touchless procurement rate >60% — PO cycle time <24hr
✓ Category strategies use market analysis + should-cost models	✓ Team trained in AI tools, negotiation, TCO, risk management
✓ Centralised contract repository with automated renewal alerts	

PROCUREMENT = STRATEGIC BUSINESS ENGINE · The Best CPOs Don't Just Track Spend — They Track Impact

Cost Intelligence	Risk Resilience	AI Automation	ESG Leadership	Inventory Mastery
Should-cost · TCO · Price floor · Channel bypass · 30–75% savings	Dual source · AI risk flags · Geopolitical diversification · Business continuity	7-step Claude framework · Touchless PO · Demand forecast · RFQ automation	Zero-waste fuel (■38L) · LED 62% reduction · Carbon tracking · Sustainable spend	EOQ · Safety stock · ABC-XYZ · JIT-Kanban · RFID · <45 day DIO target

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END OF MASTERSHEET · 2025 EDITION